



SUJET STATAPP 340 - 2026-2027

Sujet proposé par : riabi arij - arij.riabi@devoteam.com

Organisme : Devoteam SAS

Catégorie : Traitement du signal / NLP

Sujet :

Evaluating Multilingual Reasoning in Language Models

Description du sujet :

Large language models are increasingly used in multilingual settings, but reasoning evaluation remains dominated by English. Recent work shows that the same underlying reasoning problem can yield different outcomes depending on the language used. This project treats multilingual reasoning as a statistical robustness problem and exploits parallel benchmark items to separate language effects from intrinsic question difficulty.

Research question: How stable are LLM reasoning abilities across languages, and which linguistic, model-related, and prompting factors explain cross-lingual performance differences?

Données du sujet :

Primary benchmark: MMLU-ProX

MMLU-ProX is a multilingual extension of MMLU-Pro covering 29 typologically diverse languages. Each language version contains the same underlying questions, which enables paired cross-lingual comparisons. The full benchmark contains 11,829 questions per language; the Lite version contains 658 questions per language and is appropriate for a low-compute student project.

Paper: <https://aclanthology.org/2025.emnlp-main.79/>

Dataset: <https://huggingface.co/datasets/li-lab/MMLU-ProX>

Code: <https://github.com/weihao1115/MMLU-ProX>

Recommended scope: use a balanced sample of 6-8 languages spanning different scripts and resource levels, the same item IDs across languages, one or two models, and two or three prompting conditions. This preserves a paired design while keeping inference costs manageable.

Optional benchmark: MGSM

MGSM contains 250 grade-school mathematics problems manually translated into ten languages. It can be used as a smaller reproduction experiment or as an external robustness check focused specifically on mathematical reasoning.

Paper: <https://openreview.net/pdf?id=fR3wGCK-IXp>

Modèles envisagés du sujet :

Open-weight instruction-tuned models that can run on modest GPU resources or Google Colab. Suitable families include small Qwen and Llama instruction-tuned models; if resources permit, one multilingual/general model and one reasoning-oriented model can be compared. No full-model fine-tuning is required.

Bibliographie du sujet :

Xuan, W., et al. (2025). MMLU-ProX: A Multilingual Benchmark for Advanced Large Language Model Evaluation. EMNLP 2025. <https://aclanthology.org/2025.emnlp-main.79/>

Shi, F., et al. (2023). Language Models are Multilingual Chain-of-Thought Reasoners. ICLR 2023. <https://openreview.net/pdf?id=fR3wGCK-IXp>

Zhao, R., Liu, Y., Schuetze, H., & Hedderich, M. A. (2026). A Comprehensive Evaluation of Multilingual Chain-of-Thought Reasoning: Performance, Consistency, and Faithfulness Across Languages. Findings of EAACL 2026. <https://aclanthology.org/2026.findings-eaACL.276/>

Ghosh, A., Datta, D., Saha, S., & Agarwal, C. (2025). A Survey of Multilingual Reasoning in Language Models. Findings

of EMNLP 2025. <https://aclanthology.org/2025.findings-emnlp.474/>

Notation du sujet :

1

Commentaires :

LLM hors champ ENSAE en 2A.